

GreenCity

Urban Tree Canopy Monitor

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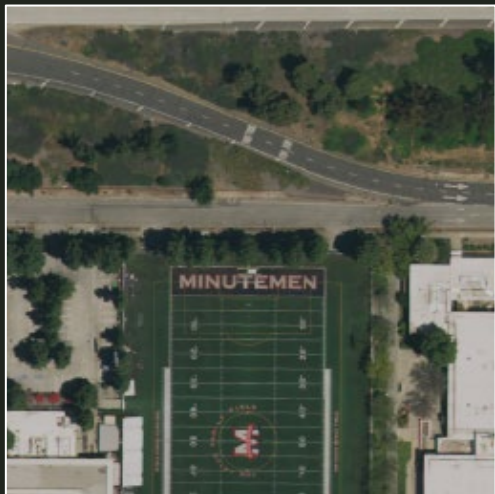


Dataset Quality

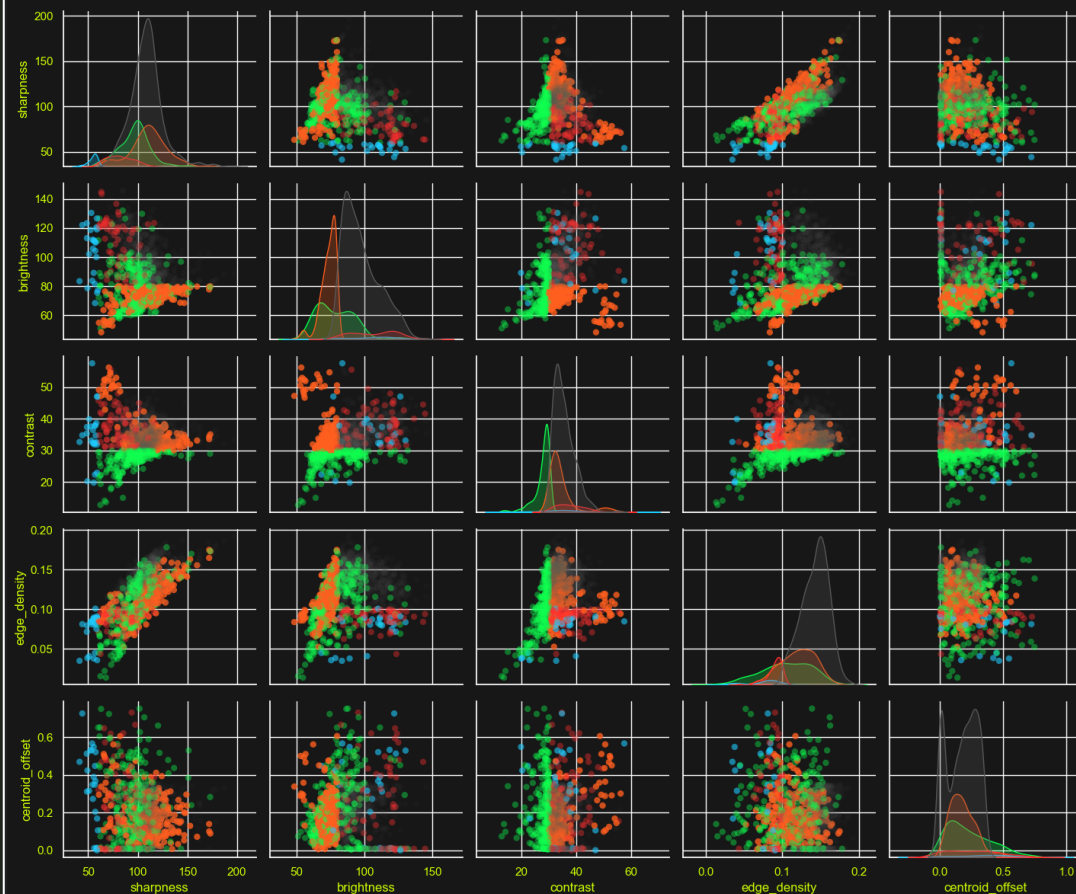
- Assessed 57K image chips across 5 CV metrics
- Metrics selected based on object detection research
- Targeted, not blanket enhancements
- Used as filters to identify poor vs. strong chips

Chip Quality Failure Modes

- low_centroid_offset
- low_edge_density
- low_sharpness
- low_brightness
- low_contrast



RAW Images: Metrics by Filter Status

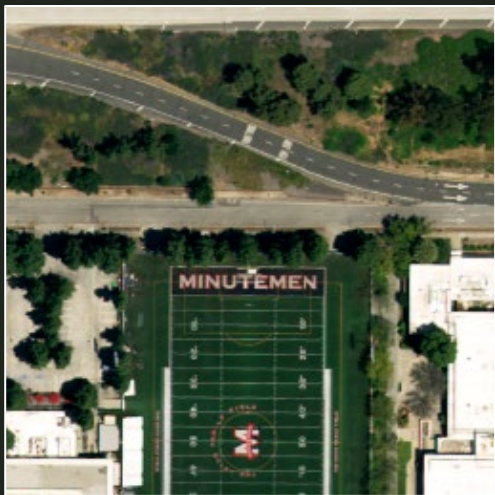


CV Rationale

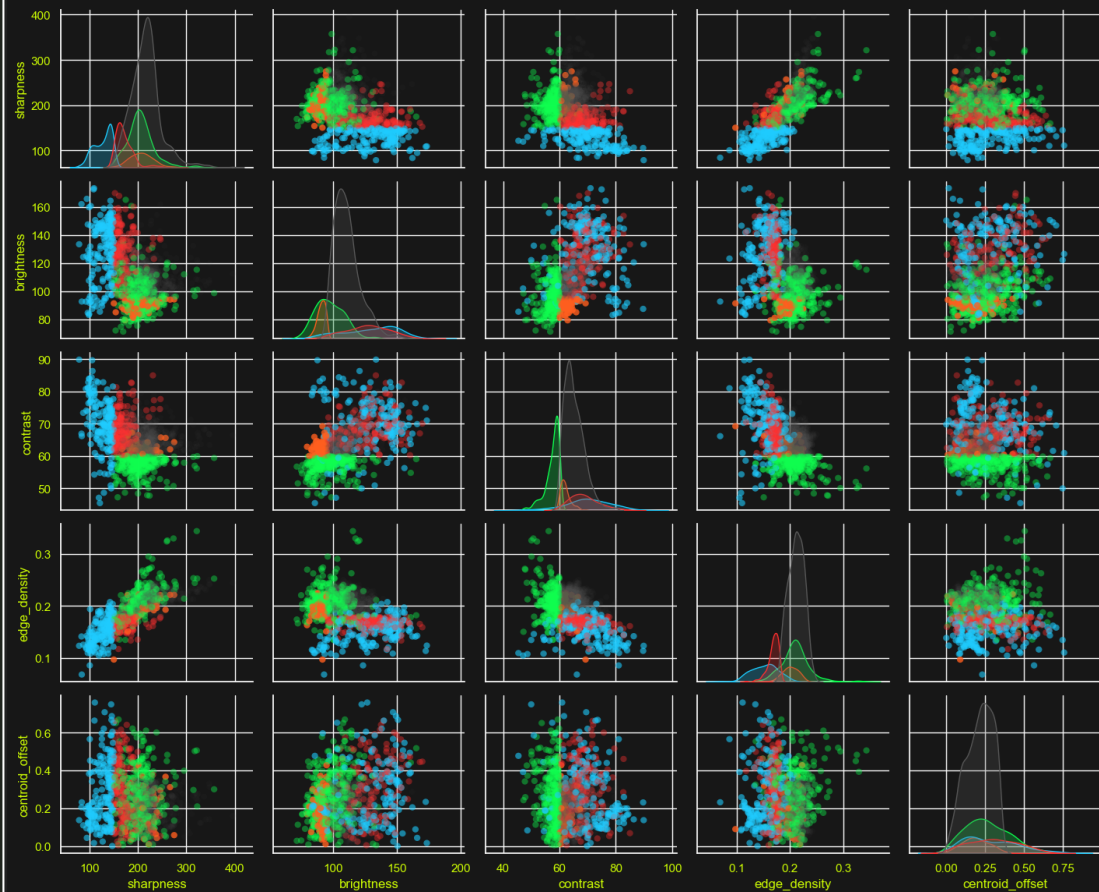
- Enhancements targeted dense/ambiguous canopy zones
- Focused on contrast and brightness
- Validated with image metrics, not just visual checks
- Improvements were measurable and task - specific

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Corrected Images: Metrics by Filter Status



Experiment Summary Table — What We Tried and Learned

Experiment	Imagery	Weights Used	Detection recall within 12 meters	Notes
1. Released model on 60cm imagery	60cm (2024)	DeepForest v1.5.2	94.02	Boxes were oversized inflating coverage by over predicting.
2. Released model on 10cm imagery	10cm (2014)	DeepForest v1.5.2	93.22%	Better, but outdated imagery.
3. Custom - trained on 60cm	60cm (2024)	Trained from scratch	80.56%	More consistent with ground truth
4. CV - enhanced imagery (trained)	60cm (2024)	Retrained & tuned	88.70%	Best so far — enhanced contrast & tuning